

NeuroAgent

A Team of AI Specialists for Reviewing Epilepsy EEG Data

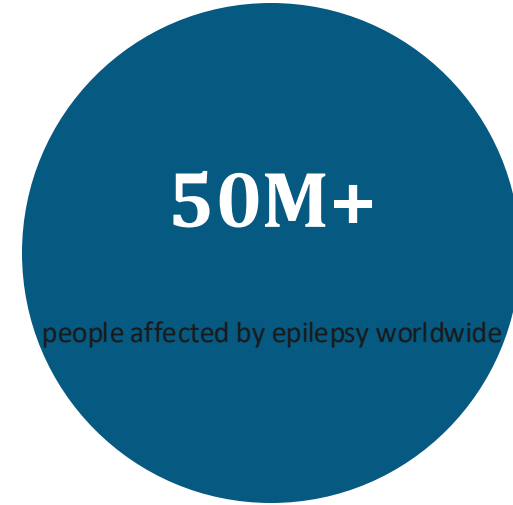
AI in Healthcare — High-Risk Project

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The Problem

Reading an EEG to figure out a patient's seizure type is hard, even for trained specialists. Two doctors can look at the same recording and disagree.

Seizure type matters because it changes what treatment works. Getting it wrong, or staying uncertain too long, has a real cost for patients.



In hospitals, doctors sometimes handle uncertain cases by asking a colleague, or a full team, to review it together. This project asks: can a small team of AI models do something similar?

The Idea: A Board of AI Specialists

Instead of one AI model giving one confident answer, NeuroAgent asks three AI models, each playing a different clinical role, to review the same EEG data independently.

Neurologist

General seizure differential

Epileptologist

Deeper reasoning on hard, ambiguous cases

Neurophysiologist

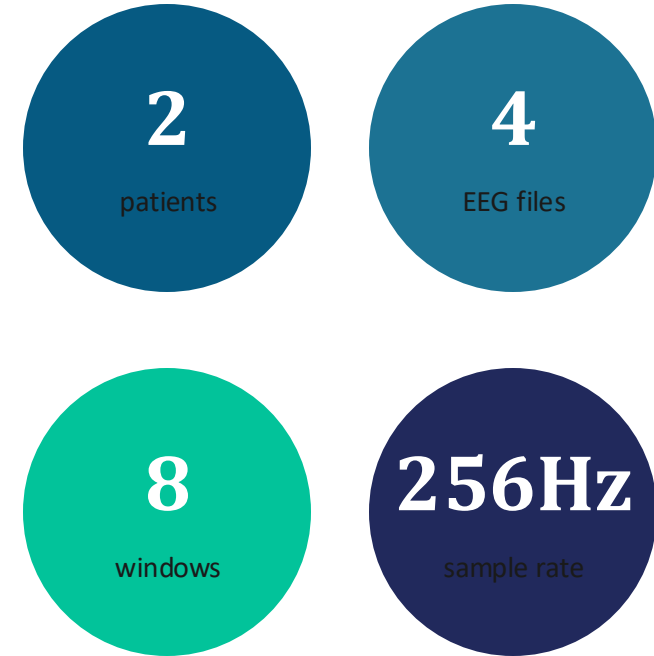
Focuses on the raw signal pattern

A fourth AI agent, the moderator, compares the three opinions and explains where they agree or disagree - instead of just picking a winner.

The Data

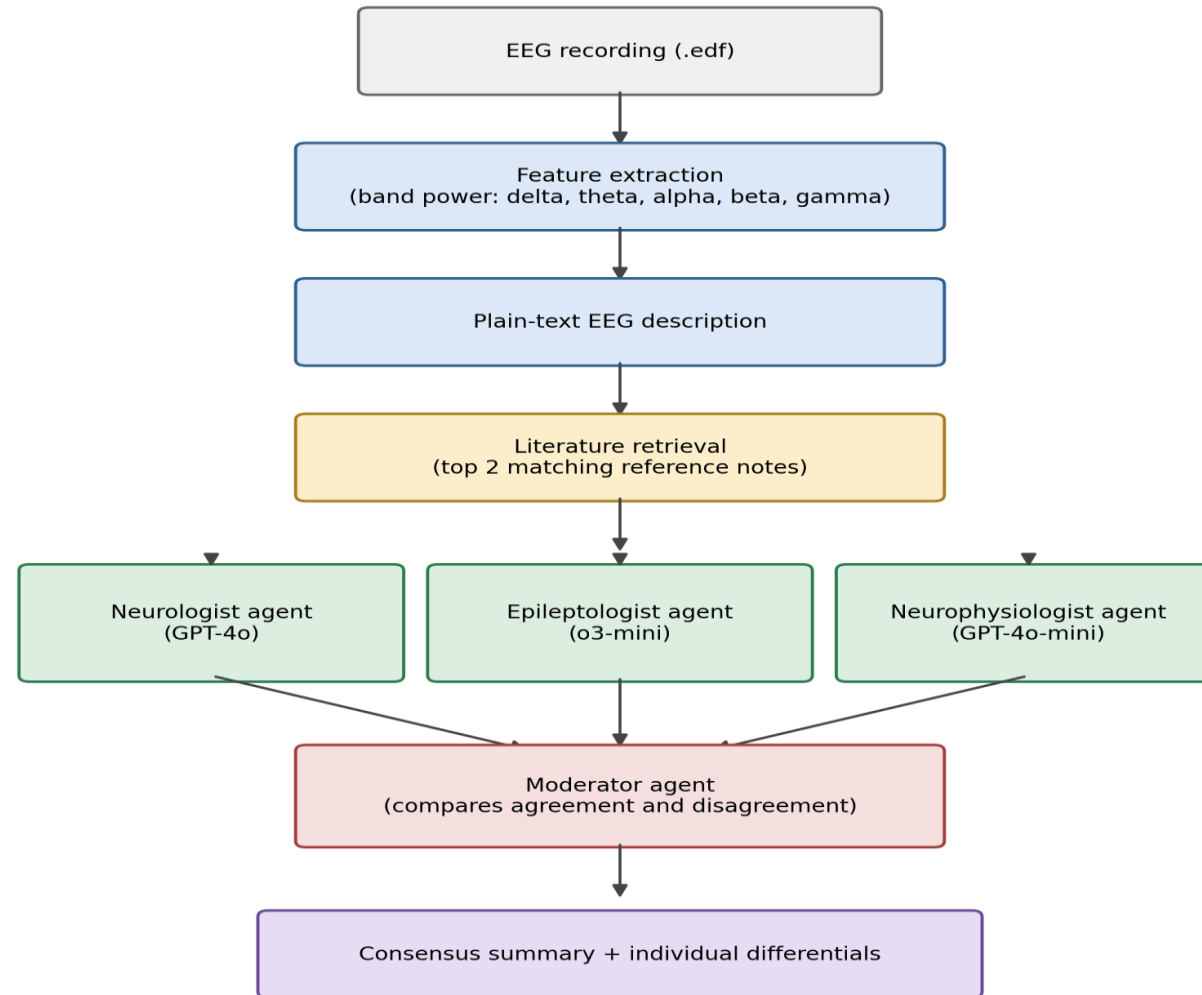
This project uses the CHB-MIT Scalp EEG Database, a public dataset of EEG recordings from pediatric epilepsy patients, hosted on PhysioNet.

From this dataset, two patient cases and four EEG files were used, giving eight 20-second EEG windows — four during a labeled seizure, four during normal activity.



How It Works

Raw EEG signal is converted into a plain-text summary of frequency patterns, matched against a small reference library, then reviewed independently by the three specialist agents before the moderator summarizes the discussion.



A Simple Comparison Point

Before trusting a complex multi-agent system, it's worth asking: how far can a simple classical model get using the same inputs?

A logistic regression model was trained on the same five frequency-band numbers the AI agents see.



75%
Accuracy

Key Finding: A Shared Blind Spot

One EEG window was mislabeled by two very different methods — for the same likely reason.

The Baseline Model

On a non-seizure window, the logistic regression baseline incorrectly predicted “seizure.”

The Epileptologist Agent

On the same window, the epileptologist agent independently leaned toward “possible ictal activity” at medium confidence — the same wrong direction.

This points to a real limitation in the input features, not a random error: the window likely reflects natural sleep activity, which looks similar to seizure activity.

What This Tells Us

Across all eight windows, the three AI specialists didn't always agree - and the moderator consistently pointed to a specific, identifiable reason: whether strong slow-wave activity should be read as sleep or as seizure.

Limitations

- All three agents use models from the same company
- No seizure-type labels yet - detection only
- Only 2 patients, 8 windows - illustrative, not validated
- No clinician has reviewed the outputs

Future Directions

Real seizure-type labels

Use a dataset like Siena or TUSZ, which have real clinician-diagnosed seizure types, not just detection labels.

True model diversity

Add agents built on models from more than one company for genuinely independent opinions.

A medication agent

A fourth agent grounded in published treatment guidelines, once seizure type is known.

Rebuild with Agent framework

Use a framework built specifically for multi-agent conversation instead of orchestration in the code.

Conclusion

NeuroAgent shows that a small team of AI specialists can surface genuine, explainable disagreement in EEG review - and that a shared blind spot between an AI agent and a classical model can point to a real gap in the data, not just noise.

It's a research prototype, and not a replacement for a real doctor's judgment.

Code submitted as a Colab notebook alongside this presentation

Thank you